

M2-Thesis

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August 2026

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Preface

Throughout my undergraduate and graduate studies in mathematics I have been caught between two main interests: the applied world of machine learning, and the theoretical world of topology and geometry. While this is often viewed as an unusual mix to choose, I have always found it greatly enjoyable.

This resulted in me wanting to answer the question of what is happening to the “shape” of data throughout a machine learning process, and also what even is the “shape” of data.

The value in understanding this is twofold: firstly, to be able to make better models by leveraging our understanding of how they work; and second, to be able to understand how to interpret what these models are doing. The dance between naturally arising data and the model trying to fit without knowing how the data is generated is fascinating. My main motivation was to try to understand this.

One big idea is the notion of data lying on some low-dimensional space, which makes both learning and sampling from distributions supported on this lower-dimensional space easier. However this belief only truly simplifies the system in the small noise regime. All sorts of complex geometry can happen in the higher noise regimes. We do not, as we will see, have a nicely behaved interpolating family of manifolds.

Before doing the IASD M2 program, I had written a master’s dissertation on diffusion models under the manifold hypothesis, under the supervision of George Deligiannidis. During this I became deeply interested in dimension estimation of datasets. This at best vague and at worst terribly ill-posed question seemed to me one where an answer would tell us a lot about the natural geometry of data, and how diffusion models are able to make use of this. Almost every paper I found on the interplay between manifolds and diffusion models cited work from Eddie Aamari.

In November 2025 I reached out to Eddie Aamari to further discuss what the geometry of data could look like, and this led to this internship project. The initial goal was to understand the interplay between geometry affecting sampling error and how this could encourage simpler embeddings in a latent diffusion setting. However, given the complexity of work being done limited time was spent on investigating auto-encoders so far.

A key challenge of this project has been striking the balance between geometry: where things are well defined, and diffusion of probability densities: where simply by adding noise we violently change the notions of geometry we would like to equip our density with. This trade-off has led to looking at several objects, which while interesting in their own right have not been helpful in bounding sampling error. I do believe that the final part on posterior covariance fields [section 6](#) does provide a new insight on where true complexity lies in the diffusion process.

Abstract

Diffusion based generative models have become ubiquitous over recent years. Their success is often attributed to the geometric structure of natural data. This is commonly referred to as the manifold hypothesis. In this report we summarise our main work so far on understanding the geometry of diffusion. We examine an anisotropic stability bound, level sets, ridges, and functionals of the posterior covariance. Ultimately, we settle on the spectrum of the posterior covariance being the key object to understanding the geometry throughout the diffusion process.

Algebra is the offer made by the devil to the mathematician. The devil says: ‘I will give you this powerful machine, it will answer any question you like. All you need to do is give me your soul: give up geometry and you will have this marvellous machine.’

— *Michael Atiyah*

1 Introduction

1.1 Initial Project

The progress in generative modelling over the last decade has been rapid. A key shift was the introduction of score based generative models (SGM), also known as diffusion models, which allow us to interpolate between a known distribution (usually a standard Gaussian), and a target distribution p_{data} , by training a network based on data $X \sim p_{\text{data}}$. By repeatedly adding noise to a target density, and training a denoising network, one can sample from p_{data} . Now, flow matching methods are the standard implementation of diffusion based generative models. These crucially can be viewed as a SGM.

While vast amounts of work has been done on improving the capabilities of diffusion models, the geometry of the entire diffusion process has been under explored.

Initially, this research internship aimed to investigate the geometry of both the auto-encoder and denoiser in latent diffusion models. If one can show that the geometry of a target distribution influences the estimation of scores, discretisation error, or sampling stability, then the case can be made that when the auto-encoder and diffusion model are trained jointly, that the auto-encoder learns a more geometrically regular embedding.

We spent the internship so far almost entirely looking at the geometry of diffusion.

1.2 Existing work

There exists literature on how different geometric aspects affect the diffusion process, but these are mostly asymptotic results for highly regular target densities (Benton et al., 2024), (Potapchik et al., 2024), (Liang et al., 2025).

The main challenge of the question “how does geometry affect diffusion” is answering what we mean by “geometry”. There are some issues with existing methods which make use of dimension and reach/curvature of the support of p_{data} :

- Dimension:
 - Ill-posed to assign a dimension $d < D$ to a density which is absolutely continuous with respect to $\mathbb{R}^D$
 - Dimension of p_{data} may not be constant
 - Dimension of p_{data} may not be well defined everywhere, for example at intersections
 - Only considers dimension of target, now how effective dimension might change.
- Reach, curvature:
 - Can be $0, \infty$ at intersections or corners
 - Can vary wildly along the support of p_{data}

1.3 Objects Investigated

This gap in the literature is the primary of the project. As such, a new object other than dimension or reach or curvature had to be found. Throughout we investigated several before landing on the final and most promising object: the posterior covariance.

- Pullback metric of flow $G_{t,T}(x)$
- Level sets
- Ridges
- Posterior Covariance

[!!!make a table comparing them after meeting!!!]

1.4 Main Contribution

We find that the spectrum of the posterior covariance captures geometric variation, and use functionals of the spectrum to understand geometric variation throughout the diffusion process. One of the things this allows us to do is pick a discretisation schedule to reduce the discretisation error in sampling.

2 Background

2.1 The Manifold Hypothesis

The manifold hypothesis offers an explanation for the effectiveness of machine learning algorithms on unstructured, high-dimensional data. It posits that a data distribution $X \sim p_{\text{data}}$ with $X \in \mathbb{R}^D$ lies on (or is concentrated near) a submanifold $\mathcal{M}$ of significantly lower dimension $d \ll D$. This restricts the effective degrees of freedom to d , enabling models to learn the underlying structure even in regimes where the sample size is smaller than the ambient dimension ($n < D$).

The primary motivation for the manifold hypothesis stems from the curse of dimensionality, wherein the volume of high-dimensional space grows exponentially, rendering data extremely sparse. High-dimensional spaces also exhibit counter-intuitive geometric behaviors; for instance, standard Gaussian data $\mathcal{N}(0, I_D)$ concentrates tightly within a thin spherical shell of radius $\sqrt{D}$.

The rationale behind the manifold hypothesis is that natural data distributions are typically locally continuous and governed by relatively few degrees of freedom. For instance, the space of realistic natural images constitutes a tiny fraction of the space of all possible pixel configurations.

Several variations of the manifold hypothesis relax its classical assumptions:

- **The Union of Manifolds Hypothesis** (Brown et al., 2022a,b): Argues that the classical single-manifold assumption is overly restrictive, modeling data instead as a union of multiple submanifolds of potentially varying dimensions.
- **The Stratified Space Hypothesis** (Aamari and Berenfeld, 2024): Extends this formulation further by permitting intersections between submanifolds of different dimensions.

For completeness, we review the formal definitions of a manifold and its reach.

Definition 2.1 (Manifold). A d -dimensional topological manifold $\mathcal{M}$ is a topological space that is locally homeomorphic to an open subset of $\mathbb{R}^d$. Specifically, for each $y \in \mathcal{M}$, there exists an open neighborhood $U_y \subseteq \mathcal{M}$ containing y and a continuous mapping $\Phi_y : U_y \rightarrow \mathbb{R}^d$ such that Φ_y is a homeomorphism onto its image. The smoothness of $\mathcal{M}$ is governed by the differentiability of the coordinate maps Φ_y .

When a manifold $\mathcal{M}$ is embedded in $\mathbb{R}^D$, its regularity can be quantified via the concept of *reach* (Federer, 1959):

Definition 2.2 (Reach). The reach $\tau = \tau(\mathcal{M})$ of an embedded submanifold $\mathcal{M} \subset \mathbb{R}^D$ is defined as

$$\tau := \sup \{ \varepsilon > 0 : \forall x \in \mathcal{M}^\varepsilon, \exists! y \in \mathcal{M} \text{ s.t. } \text{dist}(x, \mathcal{M}) = \|x - y\| \},$$

where $\mathcal{M}^\varepsilon = \{x \in \mathbb{R}^D : \text{dist}(x, \mathcal{M}) < \varepsilon\}$ denotes the open ε -neighborhood of $\mathcal{M}$. Equivalently, τ is the supremum over all neighborhood radii within which every point admits a unique orthogonal projection onto $\mathcal{M}$.

2.2 Flow Matching as a Score-Based Generative Model

Score-Based Generative Models (SGMs) (Ho et al., 2020; Song and Ermon, 2019; Song et al., 2020b) formulate generative modeling by estimating the score function, defined as the gradient of the log probability density $\nabla_x \log p(x)$. As noted by Song and Ermon (2019):

“Because gradients can be ill-defined and hard to estimate when the data resides on **low-dimensional manifolds**, we perturb the data with different levels of Gaussian noise, and jointly estimate the corresponding scores, i.e., the vector fields of gradients of the perturbed data distribution for all noise levels.”

Flow Matching (Lipman et al., 2022; Albergo et al., 2025) provides a continuous-time vector field framework that unifies and generalizes SGMs. Below, we outline the connection between Gaussian continuous flows and score matching.

Gaussian Flows and Interpolation Schedule. In flow matching, the objective is to construct a continuous transport map pushing a base noise distribution $p_1 = \mathcal{N}(0, I_D)$ to a target data distribution p_0 . Let $X_0 \sim p_0$ and $X_1 = \varepsilon \sim \mathcal{N}(0, I_D)$. The linear conditional path $X_t = \varphi_t(X_0, \varepsilon)$ interpolates between data and noise according to:

$$X_t = c_t X_0 + \sigma_t \varepsilon, \quad (1)$$

where $c_t, \sigma_t \in \mathbb{R}$ are continuously differentiable schedule functions satisfying boundary conditions $c_0 = 1, \sigma_0 = 0$ and $c_1 = 0, \sigma_1 = 1$. Differentiating X_t with respect to time t yields the conditional velocity field:

$$\dot{X}_t = \dot{c}_t X_0 + \dot{\sigma}_t \varepsilon. \quad (2)$$

Vector Field Formulation. The marginal probability density p_t of X_t evolves according to the continuity equation $\partial_t p_t = -\nabla \cdot (v_t p_t)$. The marginal vector field $v_t(x)$ generating this continuous density transformation is obtained via the conditional expectation:

$$v_t(x) = \mathbb{E}[\dot{X}_t \mid X_t = x] = \dot{c}_t \mathbb{E}[X_0 \mid X_t = x] + \dot{\sigma}_t \mathbb{E}[\varepsilon \mid X_t = x]. \quad (3)$$

Derivation via Tweedie’s Formula. Since p_t corresponds to the distribution of $c_t X_0$ corrupted by additive Gaussian noise $\sigma_t \varepsilon \sim \mathcal{N}(0, \sigma_t^2 I_D)$, Tweedie’s identity yields closed-form relationships between the conditional expectations and the score function $\nabla_x \log p_t(x)$:

$$\mathbb{E}[\varepsilon \mid X_t = x] = -\sigma_t \nabla_x \log p_t(x), \quad (4)$$

$$\mathbb{E}[X_0 | X_t = x] = \frac{x + \sigma_t^2 \nabla_x \log p_t(x)}{c_t}. \quad (5)$$

Substituting Equations (4) and (5) into the marginal velocity expression (3) gives:

$$\begin{aligned} v_t(x) &= \dot{c}_t \left(\frac{x + \sigma_t^2 \nabla_x \log p_t(x)}{c_t} \right) + \dot{\sigma}_t (-\sigma_t \nabla_x \log p_t(x)) \\ &= \frac{\dot{c}_t}{c_t} x + \left(\frac{\dot{c}_t \sigma_t^2}{c_t} - \dot{\sigma}_t \sigma_t \right) \nabla_x \log p_t(x) \\ &= \frac{\dot{c}_t}{c_t} x + \sigma_t^2 \left(\frac{\dot{c}_t}{c_t} - \frac{\dot{\sigma}_t}{\sigma_t} \right) \nabla_x \log p_t(x). \end{aligned} \quad (6)$$

Equivalence of Parameterisations. Equation (6) demonstrates that the velocity field $v_t(x)$ is uniquely determined by the noise schedules c_t, σ_t and the score function $\nabla_x \log p_t(x)$. Consequently, the following target parameterisations are equivalent up to deterministic point-wise linear transformations:

$$\nabla_x \log p_t(x) \iff \mathbb{E}[\varepsilon | X_t = x] \iff \mathbb{E}[X_0 | X_t = x] \iff v_t(x). \quad (7)$$

Sampling via the deterministic flow ODE $\dot{x}_t = v_t(x_t)$ parameterised by a score model recovers the probability flow ODE framework (e.g., DDIM (Song et al., 2020a)).

2.3 Tweedie's 1st and 2nd Order Formulae

Consider the standard unscaled Gaussian perturbation model $X_t = X_0 + \sigma_t \varepsilon$, where $\varepsilon \sim \mathcal{N}(0, I_D)$ and $X_0 \sim p_0$. The conditional density of X_t given $X_0 = x_0$ is:

$$p_t(x | x_0) = (2\pi\sigma_t^2)^{-D/2} \exp\left(-\frac{\|x - x_0\|^2}{2\sigma_t^2}\right). \quad (8)$$

The marginal density $p_t(x)$ is given by:

$$p_t(x) = \int_{\mathbb{R}^D} p_t(x | x_0) p_0(x_0) dx_0. \quad (9)$$

2.3.1 First Derivative (Score Function)

Differentiating $p_t(x)$ under the integral sign with respect to x :

$$\nabla_x p_t(x) = -\frac{1}{\sigma_t^2} \int_{\mathbb{R}^D} (x - x_0) p_t(x | x_0) p_0(x_0) dx_0. \quad (10)$$

Dividing by $p_t(x)$ and applying Bayes' rule $p(x_0 | x) = \frac{p_t(x|x_0)p_0(x_0)}{p_t(x)}$ yields Tweedie's first-order identity:

$$\nabla_x \log p_t(x) = \frac{\nabla_x p_t(x)}{p_t(x)} = \frac{\mathbb{E}[X_0 | X_t = x] - x}{\sigma_t^2}. \quad (11)$$

2.3.2 Second Derivative (Hessian Matrix / Score Jacobian)

Expanding the Hessian matrix of the log-marginal density:

$$\nabla_x^2 \log p_t(x) = \frac{\nabla_x^2 p_t(x)}{p_t(x)} - (\nabla_x \log p_t(x)) (\nabla_x \log p_t(x))^T. \quad (12)$$

Differentiating $\nabla_x p_t(x)$ a second time yields:

$$\nabla_x^2 p_t(x) = \int_{\mathbb{R}^D} \left(-\frac{1}{\sigma_t^2} I_D + \frac{(x - x_0)(x - x_0)^T}{\sigma_t^4} \right) p_t(x | x_0) p_0(x_0) dx_0. \quad (13)$$

Dividing by $p_t(x)$:

$$\frac{\nabla_x^2 p_t(x)}{p_t(x)} = -\frac{1}{\sigma_t^2} I_D + \frac{1}{\sigma_t^4} \mathbb{E}[(x - X_0)(x - X_0)^T | X_t = x]. \quad (14)$$

2.3.3 Posterior Covariance Relation

Substituting Equations (11) and (14) into Equation (12):

$$\begin{aligned} \nabla_x^2 \log p_t(x) = & -\frac{1}{\sigma_t^2} I_D + \frac{1}{\sigma_t^4} \left(\mathbb{E}[(x - X_0)(x - X_0)^T | X_t = x] \right. \\ & \left. - \mathbb{E}[x - X_0 | X_t = x] \mathbb{E}[x - X_0 | X_t = x]^T \right). \end{aligned} \quad (15)$$

Because x is deterministic given $X_t = x$, the bracketed term reduces to the conditional posterior covariance $\text{Cov}(X_0 | X_t = x)$. Thus, Tweedie's second-order formula simplifies to:

$$\nabla_x^2 \log p_t(x) = \frac{1}{\sigma_t^4} \text{Cov}(X_0 | X_t = x) - \frac{1}{\sigma_t^2} I_D. \quad (16)$$

3 Anisotropic Stability Bound

We record a stability estimate for generator matching that tracks *which* directions of a drift error are amplified and which are dampened. The scalar one-sided Lipschitz estimate propagates every error with a single worst-case rate, making it sharp only in an isotropic sense. The estimate below replaces that scalar rate with the derivative flow of the learned dynamics.

3.1 The Deterministic Transport Mechanism

Consider two time-dependent velocity fields $v_t, \hat{v}_t : \mathbb{R}^D \rightarrow \mathbb{R}^D$ for $0 \leq t \leq T$, smooth enough in space and integrable in time such that their characteristic flows are globally defined and differentiable with respect to the initial condition:

$$\frac{d}{ds} \Phi_{t,s}(x) = v_s(\Phi_{t,s}(x)), \quad \frac{d}{ds} \hat{\Phi}_{t,s}(x) = \hat{v}_s(\hat{\Phi}_{t,s}(x)), \quad \Phi_{t,t} = \hat{\Phi}_{t,t} = \text{Id}. \quad (17)$$

We write $\delta v_t := v_t - \hat{v}_t$ for the velocity error, where v_t represents the exact field and $\hat{v}_t$ the learned field.

The estimate is constructed using the derivative of the *learned* flow, which is accessible directly from its neural network architecture:

$$\hat{J}_{t,T}(x) := D_x \hat{\Phi}_{t,T}(x), \quad \frac{d}{ds} \hat{J}_{t,s}(x) = \nabla \hat{v}_s(\hat{\Phi}_{t,s}(x)) \hat{J}_{t,s}(x), \quad \hat{J}_{t,t}(x) = \text{Id}. \quad (18)$$

Thus, $\hat{J}_{t,T}(x)h$ denotes the terminal displacement at time T produced by an infinitesimal perturbation h applied at (t, x) and transported by the learned dynamics. An error at time t is not rescaled by a uniform worst-case constant; rather, it is carried along its specific direction by $\hat{J}_{t,T}$. The following identity formalizes this mechanism.

Lemma 3.1 (Telescoping identity along characteristics). *For every initial point $x_0 \in \mathbb{R}^D$, let $X_t := \Phi_{0,t}(x_0)$ denote the exact characteristic trajectory. Then,*

$$\Phi_{0,T}(x_0) - \hat{\Phi}_{0,T}(x_0) = \int_0^T \hat{J}_{t,T}(X_t) \delta v_t(X_t) dt. \quad (19)$$

Proof. Define $F(t) := \hat{\Phi}_{t,T}(X_t)$, which tracks the exact dynamics up to time t and transitions to the learned dynamics from t to T . Its boundary values represent the fully learned and fully exact terminal points: $F(0) = \hat{\Phi}_{0,T}(x_0)$ and $F(T) = \Phi_{0,T}(x_0)$. Differentiating $F(t)$ with respect to t yields:

$$\frac{d}{dt} F(t) = \partial_t \hat{\Phi}_{t,T}(X_t) + D \hat{\Phi}_{t,T}(X_t) \dot{X}_t.$$

By initiating the learned flow an instant later, we eliminate a infinitesimal learned step, giving $\partial_t \hat{\Phi}_{t,T}(x) = -D \hat{\Phi}_{t,T}(x) \hat{v}_t(x)$. Combining this with $\dot{X}_t = v_t(X_t)$ and $D \hat{\Phi}_{t,T}(X_t) = \hat{J}_{t,T}(X_t)$, we obtain:

$$\frac{d}{dt} F(t) = \hat{J}_{t,T}(X_t) (v_t(X_t) - \hat{v}_t(X_t)) = \hat{J}_{t,T}(X_t) \delta v_t(X_t).$$

Integrating $\frac{d}{dt} F(t)$ from $t = 0$ to $t = T$ proves (19). $\square$

Identity (19) shows that the local velocity error at each instant enters the terminal gap through the derivative flow evaluated precisely along the actual error direction, avoiding worst-case scalar overestimation.

3.2 From the Telescoping Identity to the Matrix Bound

We now convert this pathwise identity into a bound on the target probability measures via optimal transport coupling. Let $X_0 \sim \rho_0$ and transport it using both velocity fields: $X_t := \Phi_{0,t}(X_0)$ and $\hat{X}_t := \hat{\Phi}_{0,t}(X_0)$, yielding terminal laws $\rho_T := \text{Law}(X_T)$ and $\hat{\rho}_T := \text{Law}(\hat{X}_T)$. Driving both flows from the same random seed X_0 forms a valid coupling between ρ_T and $\hat{\rho}_T$, giving $W_1(\rho_T, \hat{\rho}_T) \leq \mathbb{E}\|X_T - \hat{X}_T\|$. Applying Lemma 3.1 pathwise at $x_0 = X_0$ alongside the integral triangle inequality yields:

$$W_1(\rho_T, \hat{\rho}_T) \leq \int_0^T \mathbb{E} \|\hat{J}_{t,T}(X_t) \delta v_t(X_t)\| dt = \int_0^T \int_{\mathbb{R}^D} \|\hat{J}_{t,T}(x) \delta v_t(x)\| \rho_t(dx) dt, \quad (20)$$

where the final equality follows from $X_t \sim \rho_t$. It remains to express the matrix norm via its underlying quadratic form.

Theorem 3.2 (Anisotropic W_1 stability). *Define the pullback amplification metric tensor:*

$$G_{t,T}(x) := \hat{J}_{t,T}(x)^\top \hat{J}_{t,T}(x), \quad \text{such that} \quad \|\hat{J}_{t,T}(x)h\|^2 = \langle h, G_{t,T}(x)h \rangle. \quad (21)$$

Then the Wasserstein distance satisfies:

$$W_1(\rho_T, \hat{\rho}_T) \leq \int_0^T \int_{\mathbb{R}^D} \langle \delta v_t(x), G_{t,T}(x) \delta v_t(x) \rangle^{1/2} \rho_t(dx) dt. \quad (22)$$

Proof. Substituting the norm identity (21) directly into (20) completes the proof. $\square$

The matrix $G_{t,T}(x)$ represents the standard Euclidean metric at terminal time T pulled back to the tangent space at (t, x) via the learned flow. It operates at time t while measuring vector magnitudes following transport to time T . Its eigenvectors identify principal directionalities at (t, x) , and its eigenvalues correspond to squared directional stretching factors. Bound (22) is anisotropic because it explicitly evaluates $G_{t,T}$ along the error vector $\delta v_t(x)$ rather than maximising over all directions.

3.3 Recovering the One-Sided Lipschitz Estimate

The classical scalar stability estimate represents the isotropic limit of Theorem 3.2, obtained by discarding directional information. Let $S_t(x) := \frac{1}{2}(\nabla \hat{v}_t(x) + \nabla \hat{v}_t(x)^\top)$ be the symmetric Jacobian component. Assume the learned field is one-sided Lipschitz with rate ℓ_t , namely:

$$\lambda_{\max}(S_t(x)) \leq \ell_t \quad \text{for all } x \in \mathbb{R}^D. \quad (23)$$

This condition restricts directional expansion by capping the maximum separation rate between trajectories, without constraining local contractions or rotational shearing in $\nabla \hat{v}_t$.

Corollary 3.3 (Scalar one-sided Lipschitz bound). *Under assumption (23),*

$$G_{t,T}(x) \preceq \exp\left(2 \int_t^T \ell_s \, ds\right) \text{Id}, \quad (24)$$

which implies the scalar stability bound:

$$W_1(\rho_T, \hat{\rho}_T) \leq \int_0^T \exp\left(\int_t^T \ell_s \, ds\right) \|\delta v_t\|_{L^1(\rho_t)} \, dt. \quad (25)$$

Proof. Fix a direction vector h and set $z_s := \hat{J}_{t,s}(x)h$, which satisfies $\dot{z}_s = \nabla \hat{v}_s(\hat{\Phi}_{t,s}(x))z_s$ according to (18). Only the symmetric Jacobian component drives norm growth:

$$\frac{d}{ds} \|z_s\|^2 = 2 \langle z_s, S_s(\hat{\Phi}_{t,s}(x))z_s \rangle \leq 2\ell_s \|z_s\|^2.$$

Applying Grönwall's inequality yields $\|\hat{J}_{t,T}(x)h\| \leq \exp\left(\int_t^T \ell_s \, ds\right) \|h\|$, establishing matrix inequality (24). Substituting (24) into (22) and using $\langle \delta v_t, G_{t,T} \delta v_t \rangle^{1/2} \leq \exp\left(\int_t^T \ell_s \, ds\right) \|\delta v_t\|$ concludes the proof. $\square$

Corollary 3.3 illustrates that the scalar estimate relies on the Loewner order bound (24): $G_{t,T}$ is uniformly dominated by its largest eigenvalue across all directions. While this bound holds generally, it ignores the eigenstructure of $G_{t,T}$, treating strongly contracting error directions identically to maximal expansion directions. This overestimation is especially pronounced when velocity errors lie normal to a low-dimensional target manifold.

3.4 A Two-Direction Model

The gap between matrix and scalar estimates can be observed in a simple two-direction model. Consider an orthogonal decomposition $\mathbb{R}^D = E_{\parallel} \oplus E_{\perp}$ into tangent-like and normal-like subspaces. Writing $x = (x_{\parallel}, x_{\perp})$, assume a linear, block-diagonal learned velocity field that expands along $E_{\parallel}$ and contracts along $E_{\perp}$:

$$\hat{v}_t(x_{\parallel}, x_{\perp}) = (\ell_t x_{\parallel}, -\gamma_t x_{\perp}), \quad \gamma_t \geq 0. \quad (26)$$

Its Spatial Jacobian $\nabla \hat{v}_t = \text{diag}(\ell_t I_{\parallel}, -\gamma_t I_{\perp})$ is constant in space. Consequently, derivative ODE (18) integrates blockwise, yielding diagonal expressions for $\hat{J}_{t,T}$ and $G_{t,T}$:

$$\hat{J}_{t,T} = \begin{pmatrix} e^{\int_t^T \ell_s \, ds} I_{\parallel} & 0 \\ 0 & e^{-\int_t^T \gamma_s \, ds} I_{\perp} \end{pmatrix}, \quad G_{t,T} = \begin{pmatrix} e^{2 \int_t^T \ell_s \, ds} I_{\parallel} & 0 \\ 0 & e^{-2 \int_t^T \gamma_s \, ds} I_{\perp} \end{pmatrix}. \quad (27)$$

Decomposing the drift error as $\delta v_t = \delta v_t^\parallel + \delta v_t^\perp$ with $\delta v_t^\parallel \in E_\parallel$ and $\delta v_t^\perp \in E_\perp$, the matrix bound integrand in (22) separates into:

$$\langle \delta v_t, G_{t,T} \delta v_t \rangle^{1/2} = \left(e^{2 \int_t^T \ell_s ds} \|\delta v_t^\parallel\|^2 + e^{-2 \int_t^T \gamma_s ds} \|\delta v_t^\perp\|^2 \right)^{1/2}. \quad (28)$$

Tangential errors are propagated to terminal time via expansion factor $e^{\int_t^T \ell}$, whereas normal errors are suppressed by decay factor $e^{-\int_t^T \gamma}$.

The scalar estimate cannot separate these components. Its expansion rate corresponds to the maximum eigenvalue of the symmetric Jacobian:

$$\lambda_{\max}(S_t) = \max\{\ell_t, -\gamma_t\} = \ell_t \quad (\ell_t \geq 0), \quad (29)$$

forcing Corollary 3.3 to propagate *both* error components using the tangential expansion factor $e^{\int_t^T \ell}$. By replacing normal contraction $e^{-\int_t^T \gamma}$ with expansion, it overestimates normal error contributions by a factor of $e^{\int_t^T (\ell_s + \gamma_s) ds}$. In regimes near target manifolds where contraction is strong ($\gamma_t \sim (T-t)^{-1}$, making $e^{-\int_t^T \gamma}$ decay polynomially in $T-t$), this overestimate creates the illusion of persistent error accumulation where damping actually occurs.

3.5 Structural Requirements on the Drift Error

The matrix and scalar bounds require different levels of information about the velocity error. The scalar estimate (25) depends only on the spatial norm $\|\delta v_t\|_{L^1(\rho_t)}$. Conversely, the matrix bound (22) depends on the directional quadratic form $\langle \delta v_t, G_{t,T} \delta v_t \rangle$, requiring knowledge of the orientation of δv_t relative to the eigenframe of $G_{t,T}$. Because the ground-truth velocity v_t is unknown in practice, $\delta v_t = v_t - \hat{v}_t$ cannot be directly evaluated.

Although empirical observations suggest that δv_t often aligns with v_t or exhibits Gaussian fluctuations across random training runs, the statistical behavior of δv_t is generally unknown prior to training.

To construct a practical coordinate system, we define fixed projection operators (P_t onto the tangential subspace and $Q_t = I - P_t$ onto the normal subspace) generated exclusively by the geometry of the target distribution. Re-estimating the eigenframe for each learned model individually would introduce model-dependent weights and complicate comparative evaluations. Therefore, we fix P_t once using the target density geometry, measuring all model errors within this single benchmark frame. The projection P_t is derived directly from the spectral decomposition of the posterior-covariance operator A_t detailed in [section 6](#).

4 Level Sets

As we smooth data lying on (or concentrated around) a manifold, two main effects occur: smoothing in the normal direction and smoothing in the tangent direction. Smoothing in the tangent direction makes a density more uniform along the manifold, leading to better-behaved level sets. Here, we investigate how these level sets evolve under smoothing.

Let $p^\star : \mathbb{R}^D \rightarrow \mathbb{R}_+$ denote the initial density, and let p_σ denote its Gaussian-smoothed counterpart:

$$p_\sigma(x) = (G_\sigma * p^\star)(x), \quad G_\sigma(x) = \frac{1}{(2\pi\sigma^2)^{D/2}} \exp\left(-\frac{\|x\|^2}{2\sigma^2}\right). \quad (30)$$

For a regular value c , the corresponding level set is

$$\mathcal{M}_c^\sigma := \{x \in \mathbb{R}^D : p_\sigma(x) = c\}, \quad \nabla p_\sigma(x) \neq 0 \quad \text{for } x \in \mathcal{M}_c^\sigma. \quad (31)$$

The unit normal vector field to $\mathcal{M}_c^\sigma$ is

$$n(x) = \frac{\nabla p_\sigma(x)}{\|\nabla p_\sigma(x)\|}. \quad (32)$$

For a tangent vector $v \in T_x \mathcal{M}_c^\sigma$, the directional derivative of the unit normal is

$$D_v n = \frac{1}{\|\nabla p_\sigma\|} (\text{Id} - nn^\top) D^2 p_\sigma v = \frac{1}{\|\nabla p_\sigma\|} P_T D^2 p_\sigma v, \quad (33)$$

where

$$P_T := \text{Id} - nn^\top \quad (34)$$

denotes the orthogonal projection onto the tangent space $T_x \mathcal{M}_c^\sigma$. Thus, the variation of the normal vector field—and consequently the local curvature of the level set—is determined by the Hessian of the smoothed density.

Gaussian smoothing regularizes not only the density itself, but all of its spatial derivatives. In particular, for any $k \geq 0$,

$$D^k p_\sigma = D^k (G_\sigma * p^\star) = (D^k G_\sigma) * p^\star. \quad (35)$$

Hence, increasing σ smooths the density while simultaneously suppressing fine-scale variations in its gradient, Hessian, and higher-order derivatives.

This smoothing has a straightforward interpretation in the Fourier domain. Let $\hat{p}_\sigma$ denote the Fourier transform:

$$\hat{p}_\sigma(\xi) = \hat{G}_\sigma(\xi) \hat{p}^\star(\xi) = \exp\left(-\frac{\sigma^2 \|\xi\|^2}{2}\right) \hat{p}^\star(\xi). \quad (36)$$

A Fourier mode of spatial frequency $\|\xi\|$ is attenuated by $\exp(-\sigma^2 \|\xi\|^2/2)$. Because high-frequency components decay exponentially with $\|\xi\|^2$, Gaussian smoothing preferentially eliminates the fine-scale spatial variations of $p^\star$.

The effect on the geometry of a level set follows directly from (33). Up to a choice of orientation for the normal vector, its shape operator is

$$S := -P_T Dn P_T = -\frac{1}{\|\nabla p_\sigma\|} P_T D^2 p_\sigma P_T. \quad (37)$$

The eigenvalues of S correspond to the principal curvatures of $\mathcal{M}_c^\sigma$. Therefore, Gaussian smoothing influences level-set geometry through two main factors: the Hessian $D^2 p_\sigma$, which controls normal variation, and $\|\nabla p_\sigma\|$, which scales density variation relative to geometric variation. Because Gaussian smoothing suppresses high-frequency components of $D^2 p_\sigma$, it dampens fine-scale variations of the shape operator and removes small-scale geometric structures from the level set.

It is important to distinguish this suppression of fine-scale variation from a uniform, pointwise reduction in curvature. In general, Gaussian smoothing does not guarantee that the principal curvatures of a fixed level set decrease monotonically with σ . Indeed, the shape operator

$$S_\sigma(x) = -\frac{1}{\|\nabla p_\sigma(x)\|} P_T D^2 p_\sigma(x) P_T, \quad x \in \mathcal{M}_c^\sigma, \quad (38)$$

depends simultaneously on $D^2 p_\sigma$ and ∇p_σ , and is evaluated along the evolving level set $\mathcal{M}_c^\sigma$. Although Gaussian smoothing attenuates high-frequency modes of $D^2 p_\sigma$, it also alters ∇p_σ and the spatial location of the level set itself. Consequently, S_σ and its eigenvalues do not satisfy a general pointwise monotonicity property.

Furthermore, the topology of $\mathcal{M}_c^\sigma$ can change as σ varies. Such topological transitions occur when the level value c passes through a critical point where

$$p_\sigma(x) = c, \quad \nabla p_\sigma(x) = 0. \quad (39)$$

For instance, two distinct connected components of a super-level set may merge through a neck as σ increases. Thus, while smoothing removes high-frequency spatial variations from the density and its derivatives, it should not be viewed as causing a monotone decrease in curvature for every individual level set. Rather, Gaussian smoothing preferentially suppresses spatial frequencies at scales smaller than σ , thereby filtering out fine-scale geometric structures encoded by the higher-order derivatives of the density.

More generally, adding Gaussian noise monotonically suppresses all L^2 Sobolev energies of the smoothed density. Let

$$p_\sigma = G_\sigma * p^*, \quad G_\sigma(x) = \frac{1}{(2\pi\sigma^2)^{D/2}} \exp\left(-\frac{\|x\|^2}{2\sigma^2}\right), \quad (40)$$

where D denotes the ambient dimension. Since p_σ satisfies the heat equation $\partial_{\sigma^2} p_\sigma = \frac{1}{2} \Delta p_\sigma$, we have, for every integer $k \geq 0$,

$$\frac{\partial}{\partial \sigma^2} \frac{1}{2} \int_{\mathbb{R}^D} |D^k p_\sigma(x)|^2 dx = -\frac{1}{2} \int_{\mathbb{R}^D} |\nabla D^k p_\sigma(x)|^2 dx \leq 0. \quad (41)$$

Increasing the noise level σ monotonically decreases the L^2 norms of all spatial derivatives of p_σ . In particular,

$$\frac{\partial}{\partial \sigma^2} \frac{1}{2} \int_{\mathbb{R}^D} |\nabla p_\sigma|^2 dx = -\frac{1}{2} \int_{\mathbb{R}^D} |\Delta p_\sigma|^2 dx \leq 0, \quad (42)$$

$$\frac{\partial}{\partial \sigma^2} \frac{1}{2} \int_{\mathbb{R}^D} |D^2 p_\sigma|^2 dx = -\frac{1}{2} \int_{\mathbb{R}^D} |\nabla \Delta p_\sigma|^2 dx \leq 0. \quad (43)$$

Equivalently, in the Fourier domain:

$$\frac{1}{2} \int_{\mathbb{R}^D} |D^k p_\sigma(x)|^2 dx \propto \frac{1}{2} \int_{\mathbb{R}^D} \|\xi\|^{2k} e^{-\sigma^2 \|\xi\|^2} |\widehat{p^\star}(\xi)|^2 d\xi. \quad (44)$$

Increasing σ progressively filters out fine-scale features, suppressing spatial frequencies $\|\xi\|$ according to $e^{-\sigma^2 \|\xi\|^2}$.

This behavior aligns with numerical intuition in generative modeling and sampling, where finer time steps are required near the final data distribution ($t \rightarrow 0$ or $\sigma \rightarrow 0$) to resolve higher-frequency density details. This phenomenon is especially prominent in high-dimensional image and audio signals. Analogously, JPEG compression exploits Fourier-domain (or discrete cosine) representations by discarding high-frequency components that contribute negligibly to human visual perception.

5 Ridges

5.1 Motivation

An initially counterintuitive aspect of the manifold hypothesis is that the lower-dimensional space around which perturbed data concentrates is typically not the original manifold $\mathcal{M}$. As the noise level increases, the effective kernel bandwidth grows, encompassing a broader region of the data space. Consequently, the noise-corrupted density concentrates around an evolving low-dimensional structure. This raises the question: *What is this evolving space, and how does its geometry behave over time?*

This phenomenon is evident even when considering the convolution of simple discrete distributions, such as a mixture of Gaussian-smoothed point masses: the modes of the smoothed distribution do not simply coincide with the original support.

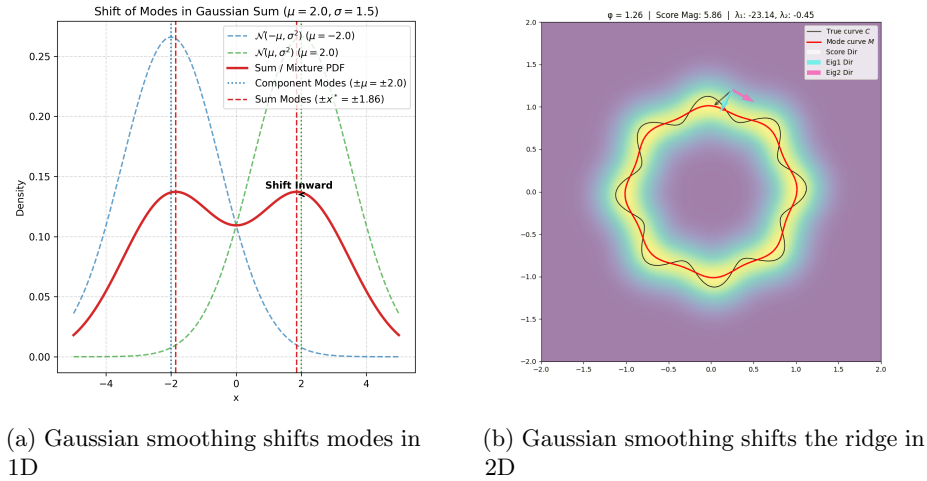


Figure 1: Comparison of ridge structures under varying smoothing scales.

With this visual intuition, we can formalize the definition of a ridge: it is the locus of points where the gradient of the density restricted to the normal directions vanishes. However, a priori, the normal directions are not fixed externally.

The key resolution is to let the density determine its own normal frame. At a point on a d -dimensional ridge, the density exhibits ridge-like behavior: it achieves a local maximum along the $D - d$ transverse directions, while varying comparatively smoothly along the remaining d intrinsic directions. Transverse to the ridge, the density curves sharply downward; along the ridge, its curvature is mild.

Mathematically, this condition is governed by the Hessian of the density.

Let $g(x) = \nabla p(x)$ and $H(x) = \nabla^2 p(x)$, and denote the eigenvalues of $H(x)$ in descending order:

$$\lambda_1(x) \geq \lambda_2(x) \geq \cdots \geq \lambda_d(x) \geq \lambda_{d+1}(x) \geq \cdots \geq \lambda_D(x), \quad (45)$$

with corresponding orthonormal eigenvectors $u_1(x), \dots, u_D(x)$. The directions of sharpest downward concavity correspond to the most negative eigenvalues. Thus, we define the normal space at x as the span of the bottom $D - d$ eigenvectors, and the tangent space as the span of the top d eigenvectors. Writing $V(x) = [u_{d+1}(x) \mid \cdots \mid u_D(x)]$ for the normal frame, the orthogonal projector onto the normal subspace is given by:

$$L(x) = V(x)V(x)^\top. \quad (46)$$

Importantly, the normal space is not specified exogenously; it is determined intrinsically by the second-order geometric structure of the density field itself.

A point x belongs to the d -dimensional ridge if the density cannot be increased by moving along any normal direction. Equivalently, the projection of the gradient onto the normal space vanishes. Defining the projected gradient:

$$G(x) := L(x)g(x), \quad (47)$$

the d -dimensional ridge set R is defined as:

$$R := \{x \in \mathbb{R}^D : \|G(x)\| = 0, \lambda_{d+1}(x) < 0\}. \quad (48)$$

The strict inequality $\lambda_{d+1}(x) < 0$ guarantees that even the largest eigenvalue in the normal direction is strictly negative, ensuring local maximality transverse to the ridge rather than a saddle point.

When $d = 0$, this definition recovers classical local modes: $L(x) = \text{Id}$, the condition $\|G(x)\| = 0$ simplifies to $\nabla p(x) = 0$, and $\lambda_1(x) < 0$ isolates local maxima. A general d -ridge relaxes this requirement by demanding critical point behavior solely within the $(D - d)$ -dimensional normal subspace, coupled with negative concavity across the normal block.

5.2 Concentration of the Score along the Ridge

We first highlight a useful structural property: the ridge of $\log p_\sigma$ and the ridge of p_σ define the identical spatial set. Because the logarithmic function is strictly monotonic, $\nabla \log p_\sigma = p_\sigma^{-1} \nabla p_\sigma$ vanishes if and only if $\nabla p_\sigma = 0$. Furthermore, their Hessians satisfy:

$$\nabla^2 \log p_\sigma = \frac{1}{p_\sigma} \nabla^2 p_\sigma - \frac{1}{p_\sigma^2} \nabla p_\sigma \nabla p_\sigma^\top, \quad (49)$$

which coincides with $p_\sigma^{-1} \nabla^2 p_\sigma$ whenever the projected gradient vanishes on the ridge.

We work primarily with the log-density $\log p_\sigma$ for two key reasons:

1. Its gradient $\nabla \log p_\sigma$ constitutes the score function, which score-based generative models directly learn.
2. As established by [Genovese et al. \(2014\)](#), the surrogate approximation property ([Theorem 5.1](#)) holds within an $O(1)$ neighborhood of $\mathcal{M}$ for the log-density, whereas it holds only within a restricted $O(\sigma)$ neighborhood for the raw density p_σ .

Theorem 5.1 (Surrogate Ridge Approximation, Theorem 7 of [\(Genovese et al., 2014\)](#)). *Consider the Hidden Manifold Model with noise variance σ^2 , and assume that the underlying manifold $\mathcal{M}$ is compact and boundaryless. Then the hyper-ridge set R_σ of the density p_σ satisfies:*

$$\text{Haus}(\mathcal{M}, R_\sigma) = O\left(\sigma^2 \log \frac{1}{\sigma}\right), \quad (50)$$

and R_σ is topologically homeomorphic to $\mathcal{M}$. Consequently, in the small-noise limit $\sigma \rightarrow 0$ with $\mathcal{M}$ fixed, the ridge R_σ converges to $\mathcal{M}$ in Hausdorff distance.

Inside an $O(\sigma)$ neighborhood around the manifold, the score operates predominantly as a normal restoring force field. [Genovese et al. \(2014\)](#) showed that:

$$\nabla \log p_\sigma(x) = -\frac{1}{\sigma^2} ((x - \hat{x}) + O(K_\sigma^2)), \quad (51)$$

where $\hat{x} = \pi_{\mathcal{M}}(x)$ denotes the orthogonal projection onto $\mathcal{M}$, and

$$K_\sigma = \sqrt{2\sigma^2 \log(\sigma^{-(A+D)})} = O\left(\sigma \sqrt{\log \frac{1}{\sigma}}\right). \quad (52)$$

The normal component of the score has magnitude $O(\sigma^{-1})$, whereas the tangential component remains $O(1)$. Thus, as $\sigma \rightarrow 0$, the score becomes a strong linear restoring force toward the ridge rather than toward $\mathcal{M}$ directly.

This distinction becomes clear by comparing first-order Taylor expansions of the score around the manifold projection $m = \pi_{\mathcal{M}}(x)$ and around the ridge projection $r = \pi_{R_\sigma}(x)$:

$$\nabla \log p_\sigma(x) = \nabla_{\mathcal{M}} \log p_\sigma(m) - \frac{1}{\sigma^2}(x - m) + O(K_\sigma^2) + o\left(\frac{\|x - m\|}{\sigma^2}\right), \quad (53)$$

$$\nabla \log p_\sigma(x) = \nabla_{R_\sigma} \log p_\sigma(r) - \frac{1}{\sigma^2}(x - r) + o\left(\frac{\|x - r\|}{\sigma^2}\right). \quad (54)$$

Expanding around the manifold leaves a residual offset of order $O(K_\sigma^2)$, whereas expanding around the ridge eliminates this term. This residual acts as a constant normal bias of magnitude $K_\sigma^2/\sigma^2 = O(\log \frac{1}{\sigma})$, which persists on the manifold $\mathcal{M}$ itself. The zero-crossing of the normal restoring field is displaced from $\mathcal{M}$ by $O(K_\sigma^2)$ along the normal direction—a displacement matching the Hausdorff distance $\text{Haus}(\mathcal{M}, R_\sigma) = O(\sigma^2 \log \frac{1}{\sigma})$ in [Theorem 5.1](#).

Projecting (54) onto the normal subspace N_r at the ridge point r formalizes this behavior. Because the intrinsic gradient $\nabla_{R_\sigma} \log p_\sigma(r)$ lies entirely within the tangent space $T_r R_\sigma$, it is annihilated by the normal projector $P_N(r)$, yielding:

$$P_N(r) \nabla \log p_\sigma(x) = -\frac{1}{\sigma^2}(x - r) + o\left(\frac{\|x - r\|}{\sigma^2}\right). \quad (55)$$

Thus, the normal score component acts as a pure linear restoring force directed toward the ridge, with universal spring constant $-\sigma^{-2}$ and no curvature-induced residual. This confirms that the ridge set constitutes the precise geometric attractor of the score function’s dominant restoring field.

5.3 Limitations of the Ridge Formulation

The underlying density fields used to define the ridge are mathematically smooth. For any $t > 0$, the noise-smoothed density p_t is a Gaussian convolution and is therefore real-analytic. Consequently, the posterior mean $m_t(y)$, the posterior covariance field $A_t(y)$, and the score Hessian $\nabla^2 \log p_t(y) = \frac{1}{t}(A_t(y) - \text{Id})$ are all analytic in y . Their eigenvalues $\lambda_1(y) \geq \dots \geq \lambda_D(y)$ vary continuously throughout $\mathbb{R}^D$ (section 6).

However, despite the smoothness of p_t , the ridge set R_t itself does not depend continuously on the data distribution. Two distinct failure modes break the continuity of R_t :

1. **Subspace Ambiguity (Eigengap Collapse):** At points where the spectral gap vanishes ($\lambda_d - \lambda_{d+1} \rightarrow 0$), the normal projector P_N becomes ill-defined. Consequently, the normal gradient condition $\|P_N \nabla \log p_t\| = 0$ becomes ill-posed as the normal eigenframe loses its orientation.
2. **Topological Discontinuities:** At locations where $\lambda_{d+1}(y) = 0$, the concavity condition triggers abruptly, causing R_t to instantly gain or lose entire connected components.

Genovese et al. (2014) address these failure modes by proving stability bounds under an assumed lower bound $\beta > 0$ on the spectral gap. Nevertheless, the underlying instability is intrinsic to the map $(m_t, A_t) \mapsto R_t$: passing smooth analytic fields through rigid scalar thresholding functions inevitably induces spatial and temporal discontinuities.

A second structural limitation is that the ridge formulation requires fixing the intrinsic dimension $d \in \mathbb{N}$ a priori. In practice, there is no single consistent choice of d across all noise scales. While dimension is a required integer parameter for normal frame projection, the natural scale-dependent effective dimension:

$$d_{\text{eff}}(t) = \mathbb{E}[\text{tr } A_t(X_t)] \quad (56)$$

is neither integer-valued nor constant in t . Instead, $d_{\text{eff}}(t)$ continuously interpolates between the intrinsic dimension d at small noise scales and zero at large noise scales, often overshooting non-monotonically at intermediate scales (section 6).

Example: Discontinuity of the Ridge Set. Consider a probability distribution supported on the unit circle $\mathbb{S}^1 \subset \mathbb{R}^2$, formed by a 1 : 3 mixture of uniform densities along the left and right semicircles, respectively. As shown in Figure 2, a minute increase in the smoothing parameter σ causes a drastic shift in both the spatial position and topology of the ridge. Furthermore, knowing the geometry of R_σ provides no information regarding how mass is distributed along it; the probability density may be negligible over large portions of the ridge set.

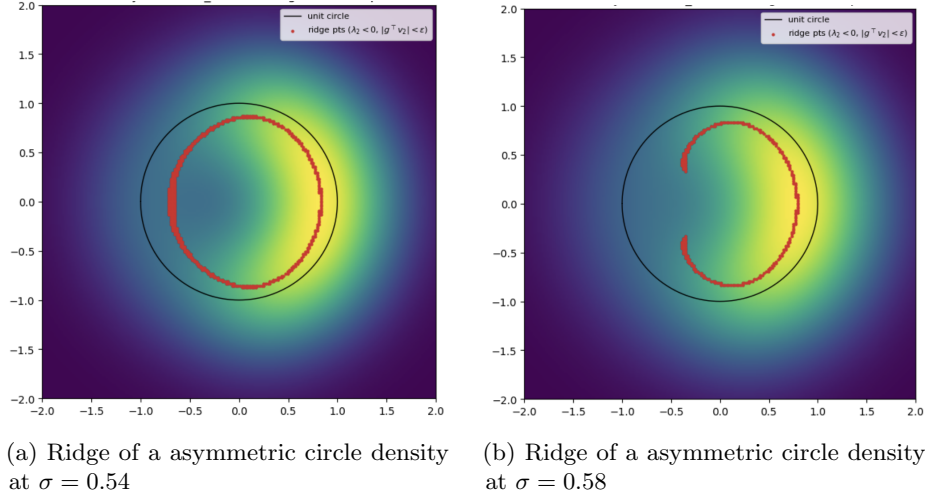


Figure 2: Comparison of ridge structures for an asymmetric semicircle density mixture at different noise levels σ . A small change in σ alters the topological structure of the ridge.

In summary, while the ridge set R_σ provides a useful theoretical surrogate for $\mathcal{M}$ at small noise levels σ and serves as the local attractor of the score field, it becomes ill-behaved and discontinuous at larger noise scales. Recognising that the ridge is fundamentally a functional of the Jacobian spectrum of the score leads naturally to the continuous formulation presented in section 6.

6 Posterior Covariance Field

Here we introduce the ultimate object we analyse: posterior covariance fields.

Definition 6.1. Let $Y_t = X + \sqrt{t}Z$, $m_t(y) := \mathbb{E}[X \mid Y_t = y]$, and

$$A_t(y) := \frac{1}{t} \text{Cov}(X \mid Y_t = y).$$

We call the weighted object $\mathfrak{F}_t := \text{Law}(Y_t, m_t(Y_t), A_t(Y_t))$ the *posterior covariance field*.

There are a few reasons why it makes sense to look at this object. First, it is not plagued by the issues we encountered in the analysis of ridges or level sets in [section 5](#) and [section 4](#). These objects suffered from being too geometric for our use case, where we care about noise scales much greater than 0. We need a more analytic quantity, and the posterior covariance field is defined for all noise scales $t > 0$. In fact level sets and ridges can just be viewed as functionals of $\mathfrak{F}_t$.

Second, the posterior covariance captures the local geometry of the score. The posterior covariance corresponds to the Jacobian of the score. This link is given in [section 2](#). This not at all obvious fact is where we reap the rewards. Here we normalise the posterior covariance by t to account for the scale of noise added.

Third, here we deal with expectations, rather than L^∞ -type quantities like the reach τ_M . If there exists a region of the density which is geometrically complex and as a consequence has poor regularity, but which carries almost no mass, we should not let it dramatically change our understanding of the behaviour of the score.

6.1 Three functionals of the spectrum

All three quantities below are moments of the same object: the spectral measure of A_t , averaged over $y \sim p_t$. Write $a_1(y), \dots, a_D(y)$ for the eigenvalues of $A_t(y)$ and let $\mu_t := \text{Law}(a_I(Y_t))$ with I uniform on $\{1, \dots, D\}$, so that $\mathbb{E} \sum_i f(a_i) = D \int f d\mu_t$ for any f .

6.1.1 Effective dimension

Definition 6.2.

$$d_{\text{eff}}(t) := \mathbb{E}[\text{tr } A_t(Y_t)] = \mathbb{E} \sum_i a_i. \quad (57)$$

Proposition 6.3 (Exact MMSE identity). *Let $\text{MMSE}(t) := \mathbb{E}|X - m_t(Y_t)|^2$. Then*

$$d_{\text{eff}}(t) = \frac{\text{MMSE}(t)}{t}. \quad (58)$$

Proof. $\text{tr Cov}(X \mid Y_t) = \mathbb{E}[|X - m_t(Y_t)|^2 \mid Y_t]$; take expectations and divide by t . $\square$

Note in identity (58) $\text{MMSE}(t)$ is precisely the population value of the ε -prediction (or x_0 -prediction) denoising loss at noise level t , so it requires no computation beyond the training objective already being evaluated.

Link to dimension. An orthogonal projector of rank k has spectrum $\{1^{(k)}, 0^{(D-k)}\}$ and trace k . In the ideal manifold limit $A_t \approx P_{T_x \mathcal{M}}$ and hence $d_{\text{eff}}(t) \approx d$. Equivalently, via (58): a d -dimensional manifold leaves $O(t)$ posterior uncertainty in each of d tangent directions and $o(t)$ in the normal directions, so $\text{MMSE} \approx d \times t$. Each eigenvalue may be read as a fractional dimension: $a_i = 0.4$ contributes 0.4 effective directions.

Failure mode. d_{eff} is a first moment and therefore cannot distinguish $\text{diag}(1, 1, 1, 0, 0, 0)$ from $\text{diag}(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$: both have trace 3. It also overcounts when $a_i > 1$, in which case the “fractional dimension” reading breaks down entirely. The next two functionals repair these two deficiencies.

6.1.2 Distance to projector

Definition 6.4.

$$\Delta_{\text{proj}}(t) := \mathbb{E} \sum_i \min\{a_i^2, (a_i - 1)^2\}. \quad (59)$$

The summand is the squared distance from a_i to the set $\{0, 1\}$, and the two branches exchange at $a_i = \frac{1}{2}$:

$$\min\{a^2, (a - 1)^2\} = \begin{cases} a^2, & a \leq \frac{1}{2} \quad (\text{read as normal-like}), \\ (a - 1)^2, & a \geq \frac{1}{2} \quad (\text{read as tangent-like}). \end{cases} \quad (60)$$

Proposition 6.5. Let $\mathcal{P} = \{P \in \mathbb{R}^{D \times D} : P = P^\top = P^2\}$ be the set of orthogonal projectors and let A be symmetric with eigenvalues a_i . Then

$$\min_{P \in \mathcal{P}} \|A - P\|_F^2 = \sum_i \min\{a_i^2, (a_i - 1)^2\}, \quad \text{hence} \quad \Delta_{\text{proj}}(t) = \mathbb{E}[\text{dist}_F^2(A_t(Y_t), \mathcal{P})]. \quad (61)$$

Proof. $\mathcal{P}$ is invariant under conjugation by orthogonal matrices, and for symmetric A the Frobenius distance to a conjugation-invariant set is minimised by a P sharing the eigenvectors of A (a consequence of the Hoffman–Wielandt inequality). It then remains to round each a_i to the nearer of 0 or 1. $\square$

Thus Δ_{proj} measures how far, on average, is the soft posterior geometry from being a genuine tangent-space projector. It vanishes exactly when $A_t(Y_t)$ is a projector almost surely.

6.1.3 Ambiguity

Definition 6.6. With $(x)_+ := \max\{x, 0\}$,

$$\mathcal{A}(t) := \mathbb{E} \sum_i (a_i - 1)_+^2. \quad (62)$$

The motivation is dynamical rather than geometric. Let (a_i, v_i) be an eigenpair of A_t . $a_i > 1$ occurs if and only if the reverse probability-flow ODE is locally expansive along v_i . The threshold where posterior variance exceeds the injected noise variance:

$$a_i = \frac{\text{posterior variance along } v_i}{\text{noise variance } t} \quad \begin{cases} < 1 : & \text{denoising, contraction,} \\ = 1 : & \text{neutral,} \\ > 1 : & \text{amplification, expansion.} \end{cases} \quad (63)$$

The three design choices are then: threshold at 1 (only expansion is penalised), measure the *excess* $a_i - 1$ (the neutral point is 1, not 0), and square it (making the functional comparable to a squared Frobenius norm and penalising strong expansion super-linearly).

Link to ambiguity. Suppose the posterior at y is approximately $\frac{1}{2}\mathcal{N}(\mu_1, \Sigma) + \frac{1}{2}\mathcal{N}(\mu_2, \Sigma)$, corresponding to two competing clean configurations. Then

$$\text{Cov}(X \mid Y_t = y) = \Sigma + \frac{1}{4}(\mu_1 - \mu_2)(\mu_1 - \mu_2)^\top, \quad (64)$$

and the between-mode term can exceed tI along $\mu_1 - \mu_2$, giving $a_i > 1$. So $a_i > 1$ signals that the noisy observation fails to identify a single locally stable latent configuration.

Remark 6.7 (Scope of interpretation). Strictly, $a_i > 1$ is a diagnostic that posterior variance exceeds the noise scale. Multimodality is the typical, but not the only, mechanism: heavy-tailed unimodal p_0 (e.g. Laplace, Student- t) also produces $a_i > 1$. “Ambiguity” is the geometric reading of the phenomenon, not a theorem.

Remark 6.8 (Mean versus tail). $\mathcal{A}$ is an average over $y \sim p_t$ and can be small while $\sup_y a_{\max}(y)$ is large, if the expansive region carries little mass. In practice one should report both $\mathcal{A}(t)$ and a high quantile of $(a_{\max} - 1)_+$, since sampler failure is driven by trajectories that enter the expansive region, not by its measure.

6.2 Evolution law of MMSE

By [Theorem 6.9](#), differentiating $\text{MMSE}(t)$ under the Gaussian channel model $Y_t = X + \sqrt{t}Z$ gives:

$$\text{MMSE}'(t) = \frac{1}{t^2} \mathbb{E} \left[\text{tr} \left(\text{Cov}(X \mid Y_t)^2 \right) \right] \quad (65)$$

By definition, $A_t(y) = \frac{1}{t} \text{Cov}(X \mid Y_t = y)$. Squaring this relation and taking the trace yields:

$$\text{tr}(A_t(Y_t)^2) = \frac{1}{t^2} \text{tr}(\text{Cov}(X \mid Y_t)^2) = \|A_t(Y_t)\|_F^2 \quad (66)$$

Substituting this directly into the expression for $\text{MMSE}'(t)$:

$$\text{MMSE}'(t) = \mathbb{E}[\text{tr}(A_t(Y_t)^2)] = \mathbb{E}\|A_t(Y_t)\|_F^2 \quad (67)$$

6.3 MMSE-driven discretisation

We consider the additive Gaussian perturbation

$$X_t = X_0 + \sqrt{t} Z, \quad Z \sim \mathcal{N}(0, I_D), \quad (68)$$

so that, in the notation of the diffusion model,

$$c_t = 1, \quad \sigma_t^2 = t. \quad (69)$$

Let

$$m_t(x) := \mathbb{E}[X_0 \mid X_t = x] \quad (70)$$

denote the posterior mean and define the minimum mean-squared error

$$\text{MMSE}(t) := \mathbb{E}[\|X_0 - m_t(X_t)\|^2]. \quad (71)$$

Following the effective-dimension viewpoint, we write

$$d_{\text{eff}}(t) := \frac{\text{MMSE}(t)}{t}, \quad (72)$$

and hence

$$\text{MMSE}(t) = t d_{\text{eff}}(t). \quad (73)$$

Discretisation error. For additive Gaussian noise, Tweedie's formula gives

$$s_t(x) = \frac{m_t(x) - x}{t}. \quad (74)$$

The same martingale argument used in (Potapchik et al., 2024) to control the score variation in the discretised reverse process then yields, for two noise levels $t < t'$,

$$\begin{aligned} & \mathbb{E} \|s_t(X_t \mid t', X_{t'}) - s_t(X_t)\|^2 \\ &= \frac{1}{t^2} (\text{MMSE}(t') - \text{MMSE}(t)), \end{aligned} \quad (75)$$

with the sign understood according to the direction in which the diffusion trajectory is traversed. Thus, for a sufficiently fine discretisation with step size

$$\Delta t_k := t_{k+1} - t_k, \quad (76)$$

the contribution of the k -th interval is, up to higher-order terms,

$$\mathcal{E}_k \asymp \frac{|\text{MMSE}'(t_k)|}{t_k^2} (\Delta t_k)^2. \quad (77)$$

Consequently, the total discretisation error is approximated by

$$\mathcal{E}_{\text{disc}} \asymp \sum_{k=0}^{K-1} \alpha(t_k) (\Delta t_k)^2, \quad \alpha(t) := \frac{|\text{MMSE}'(t)|}{t^2}. \quad (78)$$

Optimal allocation of time steps. We now choose the discretisation points to minimise (78) subject to a fixed number K of steps. Introduce the local step density

$$n(t) := \frac{1}{\Delta t(t)}. \quad (79)$$

In the continuum limit,

$$K \approx \int n(t) dt, \quad (80)$$

while

$$\begin{aligned} \mathcal{E}_{\text{disc}} &\approx \int \alpha(t) \Delta t(t) dt \\ &= \int \frac{\alpha(t)}{n(t)} dt. \end{aligned} \quad (81)$$

Minimising this functional subject to $\int n(t) dt = K$ gives

$$n(t) \propto \sqrt{\alpha(t)}, \quad (82)$$

and therefore

$$\Delta t(t) \propto \frac{1}{\sqrt{\alpha(t)}} = \frac{t}{\sqrt{|\text{MMSE}'(t)|}}. \quad (83)$$

Finally, using Equation 67,

$$\text{MMSE}'(t) = \mathbb{E} \|A_t(Y_t)\|_F^2, \quad (84)$$

so the resulting MMSE-adaptive discretisation is

$$\Delta t(t) \propto \frac{t}{\sqrt{\mathbb{E} \|A_t(Y_t)\|_F^2}}. \quad (85)$$

6.4 Interpretation

Geometric Interpretation. We see the effect of the $\text{MMSE}'(t)$ term on time-step density assignment. Notably, since $t d_{\text{eff}}(t) = \text{MMSE}(t)$, we have $\text{MMSE}'(t) = d_{\text{eff}}(t) + t d'_{\text{eff}}(t)$. Hence for higher d_{eff} and crucially changing d_{eff} for larger time we require smaller time steps. Furthermore, $\mathbb{E} \|A_t(Y_t)\|_F^2 = \sum_i \mathbb{E} [a_i(t)^2]$, so this discretisation assigns a high density of small time-steps whenever we have expansive eigenvalues $a_i > 1$, in expectation.

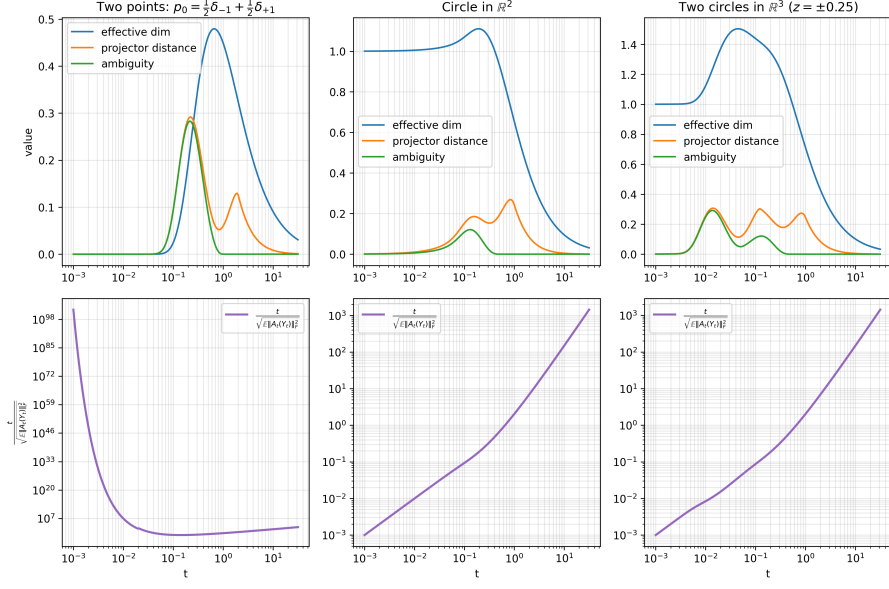


Figure 3: Functionals and assigned time step evolution for Bernoulli($\{+1, -1\}$), Uniform($\mathcal{S}^1$), and Uniform($\{\mathcal{S}^1\} \times \{\frac{1}{4}, -\frac{1}{4}\}$)

Information Theoretic Interpretation. By (Guo et al., 2005) we have

$$\frac{d}{dt}I(X_0; X_t) = \frac{dI}{d\gamma} \cdot \frac{d\gamma}{dt} = \left(\frac{1}{2} \text{MMSE}(t)\right) \left(-\frac{2}{t^3}\right) = -\frac{\text{MMSE}(t)}{t^3}$$

where γ is the signal to noise ratio (here $1/t^2$). More to be said!

Probabilistic Interpretation. To be discussed.

Acknowledgements

I am deeply grateful to my supervisors Eddie Aamari and Simon Coste, who are the sole reason this research internship was possible and helped me explore a question I had for more than a year.

AI Usage

Image to text converters were used to aid in typesetting equations, and large language models were used for some code generation and in editing parts of the report.

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Appendix: MMSE Evolution

Theorem 6.9. Let $X \in \mathbb{R}^D$ be a random vector with probability density $p(x)$ and finite second moment, and let $Z \sim \mathcal{N}(0, I_D)$ be independent of X . Define the observation process $Y_t = X + \sqrt{t}Z$ for $t > 0$, and the Minimum Mean Squared Error function:

$$\text{MMSE}(t) = \mathbb{E} [\|X - \mathbb{E}[X | Y_t]\|^2]$$

Then $\text{MMSE}(t)$ is differentiable with respect to $t > 0$, and its derivative satisfies:

$$\text{MMSE}'(t) = \frac{1}{t^2} \mathbb{E} [\text{tr} (\text{Cov}(X | Y_t)^2)]$$

Proof. The marginal density of Y_t , $p_t(y) = \int p(x)(2\pi t)^{-D/2} \exp\left(-\frac{\|y-x\|^2}{2t}\right) dx$, satisfies the heat equation $\frac{\partial p_t(y)}{\partial t} = \frac{1}{2} \Delta_y p_t(y)$. Define the score function $s_t(y) = \nabla_y \log p_t(y)$ and the Fisher Information trace $J(t) = \mathbb{E} [\|s_t(Y_t)\|^2]$.

1. Score representation of conditional statistics. Spatial differentiation of $p_t(y)$ yields Tweedie's formula:

$$s_t(y) = \frac{\nabla_y p_t(y)}{p_t(y)} = -\frac{y - \mathbb{E}[X | Y_t = y]}{t} \implies X - \mathbb{E}[X | Y_t] = -ts_t(Y_t)$$

Taking the spatial Jacobian of $\mathbb{E}[X | Y_t = y] = y + ts_t(y)$:

$$\nabla_y \mathbb{E}[X | Y_t = y] = \frac{1}{t} \text{Cov}(X | Y_t = y) = I_D + t \nabla_y s_t(y)$$

Rearranging yields the score Hessian:

$$\nabla_y s_t(y) = \frac{\text{Cov}(X | Y_t = y) - tI_D}{t^2}$$

2. Relation between MMSE and Fisher Information. By Tweedie's formula:

$$\text{MMSE}(t) = \mathbb{E} [\| -ts_t(Y_t) \|^2] = t^2 J(t)$$

Applying integration by parts ($\mathbb{E}[\|s_t(Y_t)\|^2] = -\mathbb{E}[\text{tr}(\nabla_y s_t(Y_t))]$) to the score Hessian:

$$\begin{aligned} J(t) &= -\mathbb{E} \left[\text{tr} \left(\frac{\text{Cov}(X | Y_t) - tI_D}{t^2} \right) \right] = \frac{D}{t} - \frac{\text{MMSE}(t)}{t^2} \\ \implies \text{MMSE}(t) &= tD - t^2 J(t) \end{aligned}$$

3. Time evolution of Fisher Information. Differentiating $J(t) = \int p_t(y) \|\nabla \log p_t(y)\|^2 dy$ with respect to t under heat flow $\frac{\partial p_t}{\partial t} = \frac{1}{2} \Delta p_t$ yields de Bruijn's score identity:

$$J'(t) = -\mathbb{E} [\|\nabla_y s_t(Y_t)\|_F^2] = -\mathbb{E} [\text{tr} ((\nabla_y s_t(Y_t))^2)]$$

Substituting the expression for $\nabla_y s_t(Y_t)$:

$$\begin{aligned} J'(t) &= -\mathbb{E} \left[\text{tr} \left(\left(\frac{\text{Cov}(X | Y_t) - t I_D}{t^2} \right)^2 \right) \right] \\ &= -\frac{1}{t^4} \mathbb{E} [\text{tr} (\text{Cov}(X | Y_t)^2 - 2t \text{Cov}(X | Y_t) + t^2 I_D)] \\ &= -\frac{1}{t^4} \mathbb{E} [\text{tr} (\text{Cov}(X | Y_t)^2)] + \frac{2}{t^3} \text{MMSE}(t) - \frac{D}{t^2} \end{aligned}$$

4. Total differentiation of MMSE(t). Differentiating $\text{MMSE}(t) = tD - t^2 J(t)$ via the product rule:

$$\text{MMSE}'(t) = D - 2tJ(t) - t^2 J'(t)$$

Substituting $2tJ(t) = 2D - \frac{2\text{MMSE}(t)}{t}$ and $-t^2 J'(t)$:

$$\begin{aligned} \text{MMSE}'(t) &= D - \left(2D - \frac{2\text{MMSE}(t)}{t} \right) + \left(\frac{1}{t^2} \mathbb{E} [\text{tr} (\text{Cov}(X | Y_t)^2)] - \frac{2\text{MMSE}(t)}{t} + D \right) \\ &= (D - 2D + D) + \left(\frac{2\text{MMSE}(t)}{t} - \frac{2\text{MMSE}(t)}{t} \right) + \frac{1}{t^2} \mathbb{E} [\text{tr} (\text{Cov}(X | Y_t)^2)] \\ &= \frac{1}{t^2} \mathbb{E} [\text{tr} (\text{Cov}(X | Y_t)^2)] \end{aligned}$$

□